Internal Audit Certificate

Paper III — Linear-Error Clique Partitions of Split Graphs (Erdos #81)

Audit block: Block 03 — Unified fractional margin (Thm 4.2 / E-4.2)

VERDICT: PASS (78,384/78,384 checks)

Scope

Exact-rational grid audit of the completion-of-squares inequality (4.5) F(p,q,d) >= q*d/2 + (C_alpha+mu(alpha))*p^2 - p/2 over 3<=p<=48, 1<=q<=2p, 0<=d<=p, plus third-branch dominance bookkeeping.

Summary of evidence

- 78,384/78,384 exact-rational margin checks PASS.
- Third (hot-neighbourhood) branch co-minimises in 36,317 cases and never violates the margin.

Methodology

All numeric checks use exact rational arithmetic (Python fractions) or exact linear/integer programming; symbolic identities are proved with SymPy (simplify(LHS-RHS)=0 or sum-of-squares certificate). The audit re-derives each claim independently of the manuscript's own scripts and of the Lean formalization.

Reproducibility

Results file: results/margin_results.txt
SHA-256 (results file): 577450739c3652ed9be447a3b426f23f8e856dd7eac4aa37002bcdb859b5239a
Generated: 2026-07-21 16:38 UTC | Tooling: Python 3.14, SymPy 1.14, SciPy 1.17, PuLP/CBC
Auditor: internal automated audit harness (Claude Code)

This certificate attests to computational/symbolic verification of the stated finite/closed-form claims at the audited parameter ranges. It complements, and is independent of, the machine-checked Lean 4 / Mathlib formalization.